

Generalized Linear Mixed Models

Model

Definition 1.1. Let Y_{ki} denote the i th response in the k th cluster. Suppose that, conditional on the covariates $(\mathbf{x}_{ki}, \mathbf{z}_{ki})$ and the **random effect** γ_k , Y_{ki} has a distribution in the exponential dispersion family. A **generalized linear mixed model** (GLMM) takes the following form:

$$g(\mu_{ki}) = \eta_{ki} \equiv \mathbf{x}'_{ki}\boldsymbol{\beta} + \mathbf{z}'_{ki}\boldsymbol{\gamma}_k. \quad (1.1)$$

Here g is the link function, $\mu_{ki} = \mathbb{E}(Y_{ki}|\boldsymbol{\gamma}_k, \mathbf{x}_{ki}, \mathbf{z}_{ki})$ is the conditional expectation of Y_{ki} , η_{ki} is the linear predictor, and $\boldsymbol{\beta}$ is described as the **fixed effect**. The random effects are taken as IID across clusters, with mean $\mathbb{E}(\boldsymbol{\gamma}_k) = \mathbf{0}$, and variance $\mathbb{V}(\boldsymbol{\gamma}_k) = \mathbf{G}(\boldsymbol{\alpha})$. Parameters $\boldsymbol{\alpha}$ for the random effect model are described as **variance components**. If a dispersion parameter ϕ is required, absorb this into $\boldsymbol{\alpha}$.

Define $\mathbf{y}_k = \text{vec}(Y_{k1}, \dots, Y_{kn_k})$ as the response vector for cluster k . Let $\mathcal{D}_k$ collect the covariates relevant to cluster k :

$$\mathcal{D}_k = \bigcup_{i=1}^{n_k} \{\mathbf{x}_{ki}, \mathbf{z}_{ki}\}.$$

Observations belonging to distinct clusters $\mathbf{y}_k$ and $\mathbf{y}_l$ are independent given covariates:

$$(\mathbf{y}_k \perp \mathbf{y}_l) | (\mathcal{D}_k, \mathcal{D}_l).$$

Within a given cluster k , observations Y_{ki} and Y_{kj} are conditionally independent given covariates and the cluster's random effect:

$$(Y_{ki} \perp Y_{kj}) | (\boldsymbol{\gamma}_k, \mathbf{x}_{ki}, \mathbf{x}_{kj}).$$

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Remark 1.1. In contrast to LMMs, GLMMs generally assume that observations within a cluster are conditionally independent given the cluster's random effect. ◆

Remark 1.2. The components of a GLMM are summarized here: ◆

Structure	Dimension	Description
β	$p \times 1$	Fixed effect.
γ_k	$q \times 1$	Random effect.
η_{ki}	1×1	Linear predictor.
α	$r \times 1$	Variance components.
$G(\alpha)$	$q \times q$	Random effect covariance.

Objective Function

2.1 Likelihood

Result 2.1 (Model Likelihood). Suppose there are K clusters of size n_k . Let $\mathcal{D}_k$ denote the covariates relevant to the k th cluster. The GLMM likelihood is:

$$L(\beta, \alpha) = \prod_{k=1}^K \int \left\{ \prod_{i=1}^{n_k} f(y_{ki} | \mathcal{D}_{ki}, \gamma_k; \beta, \alpha) \right\} f(\gamma_k | \alpha) d\gamma_k. \quad (2.1)$$

♣

Proof. Note that the random effects $\{\gamma_k\}$ are not among the observed data. The likelihood for the observed data factors as:

$$\begin{aligned} \mathcal{L}(\beta, \alpha) &\equiv f(\mathbf{y} | \mathcal{D}; \beta, \alpha) = \prod_{k=1}^K f(\mathbf{y}_k | \mathcal{D}_k; \beta, \alpha) = \prod_{k=1}^K \int f(\mathbf{y}_k, \gamma_k | \mathcal{D}_k; \beta, \alpha) d\gamma_k \\ &= \prod_{k=1}^K \int f(\mathbf{y}_k | \mathcal{D}_k, \gamma_k; \beta, \alpha) f(\gamma_k | \alpha) d\gamma_k. \end{aligned}$$

Since observations within a cluster are conditionally independent given covariates $\mathcal{D}_k$ and the cluster's random effect γ_k :

$$\prod_{k=1}^K \int f(\mathbf{y}_k | \mathcal{D}_k, \gamma_k; \beta, \alpha) f(\gamma_k | \alpha) d\gamma_k = \prod_{k=1}^K \int \left(\prod_{i=1}^{n_k} f(y_{ki} | \mathcal{D}_{ki}, \gamma_k; \beta, \alpha) \right) f(\gamma_k | \alpha) d\gamma_k.$$

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2.2 Quasi-Likelihood

Remark 2.1. To perform estimation and inference in the GLMM framework, it suffices to specify the following components:

- i. A mean model: $g(\mu_{ki}) = \eta_{ki}$.
- ii. The mean-variance relationship: $\mathbb{V}(Y_{ki}) = \phi\nu(\mu_{ki})$.

The objective function induced by these two components is the *quasi (log) likelihood*. $\blacklozenge$

Definition 2.1. The **quasi likelihood** contribution of y_{ki} is defined as:

$$\ell_{ki} = \ell(y_{ki}|\mathcal{D}_{ki}, \gamma_k; \beta, \alpha) = \int_{y_{ki}}^{\mu_{ki}} \frac{y_{ki} - u}{\phi\nu(u)} du. \quad (2.2)$$

See [3] for the definition of quasi-likelihood. $\blacksquare$

Example 2.1. From (2.1), the GLMM log likelihood is:

$$\ell(\beta, \alpha) = \sum_{k=1}^K \ln \int \exp \left\{ \sum_{i=1}^{n_k} \ln f(y_{ki}|\mathcal{D}_{ki}, \gamma_k; \beta, \alpha) \right\} f(\gamma_k|\alpha) d\gamma_k.$$

Suppose the random effects $\gamma_k \sim N(\mathbf{0}, \mathbf{G})$, then:

$$\ell(\beta, \alpha) \propto \sum_{k=1}^K \ln \int \exp \left\{ \sum_{i=1}^{n_k} \ln f(y_{ki}|\mathcal{D}_{ki}, \gamma_k; \beta, \alpha) \right\} \det(\mathbf{G})^{-1/2} e^{-\gamma_k' \mathbf{G}^{-1} \gamma_k} d\gamma_k.$$

Substituting the quasi likelihood ℓ_{ki} for $\ln f(y_{ki}|\mathcal{D}_{ki}, \gamma_k; \beta, \alpha)$ and reducing gives:

$$\ell(\beta, \alpha) \propto -\frac{K}{2} \ln \det(\mathbf{G}) - \sum_{k=1}^K \ln \int \exp \left\{ -\frac{1}{2} \gamma_k' \mathbf{G}^{-1} \gamma_k + \sum_{i=1}^{n_k} \ell_{ki} \right\} d\gamma_k \quad (2.3)$$

$\spadesuit$

2.3 Laplace Approximation

Remark 2.2. The integral appearing in (2.3) is typically intractable. In the *Laplace approximation* to the GLMM quasi likelihood, the integrand is approximated via a second order Taylor expansion, then evaluated analytically to obtain an approximate objective function. $\blacklozenge$

Proposition 2.1. Suppose $\gamma_k \stackrel{\text{IID}}{\sim} N(\mathbf{0}, \mathbf{G})$. Define $s_k : \mathbb{R}^q \rightarrow \mathbb{R}$:

$$s_k(\gamma) = \frac{1}{2} \gamma' \mathbf{G}^{-1} \gamma - \sum_{i=1}^{n_k} \ell_{ki}(\gamma).$$

Taking $\gamma_k^* = \arg \min_{\gamma} s_k(\gamma)$ as the expansion point, the Laplace approximation to the quasi likelihood is:

$$\ell_q(\beta, \alpha, \gamma^*) \propto -\frac{K}{2} \ln \det(\mathbf{G}) - \sum_{k=1}^K s_k(\gamma_k^*) - \frac{1}{2} \sum_{k=1}^K \ln \det \left\{ \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'}(\gamma_k^*) \right\}. \quad (2.4)$$

$\blacklozenge$

Proof. Define $I_k \in \mathbb{R}$ as:

$$I_k = \int \exp \left\{ -\frac{1}{2} \gamma_k' \mathbf{G}^{-1} \gamma_k + \sum_{i=1}^{n_k} \ell_{ki}(\gamma_k) \right\} d\gamma_k = \int e^{-s_k(\gamma)} d\gamma. \quad (2.5)$$

Take the 2nd order Taylor expansion of $s_k(\gamma)$ about $\gamma_k^* = \arg \min s_k(\gamma)$:

$$s_k(\gamma) = s_k(\gamma_k^*) + \frac{\partial s_k}{\partial \gamma}(\gamma_k^*) \cdot (\gamma - \gamma_k^*) + \frac{1}{2} (\gamma - \gamma_k^*) \cdot \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'}(\gamma_k^*) \cdot (\gamma - \gamma_k^*) + \mathcal{O}(\|\gamma - \gamma_k^*\|^3).$$

Since γ_k^* is the maximum of $s_k(\gamma)$, the gradient of s_k at γ_k^* vanishes:

$$\frac{\partial s_k}{\partial \gamma}(\gamma_k^*) = \mathbf{0}.$$

Substitute the Taylor expansion for $s_k(\gamma)$ into (2.5) to obtain:

$$\hat{I}_k = \int_{\gamma} \exp \left\{ -s_k(\gamma_k^*) - \frac{1}{2} (\gamma - \gamma_k^*) \cdot \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'}(\gamma_k^*) \cdot (\gamma - \gamma_k^*) \right\} d\gamma. \quad (2.6)$$

To evaluate $\hat{I}_k$, note that for $\mathbf{z}_{q \times 1} \sim N(\mathbf{0}, \Sigma)$:

$$\int_{\mathbb{R}^q} (2\pi)^{-q/2} \det(\Sigma)^{-1/2} \exp \{ -\mathbf{z}' \Sigma^{-1} \mathbf{z} \} d\mathbf{z} = 1.$$

Rearranging gives the identities:

$$\int_{\mathbb{R}^q} \exp \{ -\mathbf{z}' \Sigma^{-1} \mathbf{z} \} d\mathbf{z} = (2\pi)^{q/2} \det(\Sigma)^{1/2}.$$

Replacing Σ^{-1} by Σ :

$$\int_{\mathbb{R}^q} \exp \{ -\mathbf{z}'\Sigma\mathbf{z} \} d\mathbf{z} = (2\pi)^{q/2} \det(\Sigma)^{-1/2}.$$

Therefore, (2.6) evaluates to:

$$\hat{I}_k(\gamma_k^*) = e^{-s_k(\gamma_k^*)} (2\pi)^{q/2} \det \left\{ \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'}(\gamma_k^*) \right\}^{-1/2}.$$

Taking the logarithm:

$$\ln \hat{I}_k(\gamma_k^*) \propto -s_k(\gamma_k^*) - \frac{1}{2} \ln \det \left\{ \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'}(\gamma_k^*) \right\}.$$

Finally, the approximate log integrated quasi likelihood is:

$$\ell(\beta, \alpha) = -\frac{K}{2} \ln \det(\mathbf{G}) + \sum_{k=1}^K \ln \hat{I}_k(\gamma_k^*).$$

■

Remark 2.3. To evaluate the Laplace likelihood (2.4), it remains to:

- i. Provide an expression for the cluster-specific expansions points γ_k^* .
- ii. Provide the form of the Hessian of $s_k(\gamma)$.

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Proposition 2.2. Recall the following matrices defined in the context of GLMs:

$$\Delta_k = \text{diag}\{\dot{g}(\mu_{ki})\}, \quad \mathbf{W}_k = \text{diag}\left\{ \frac{1}{\phi\nu(\mu_{ki})\dot{g}^2(\mu_{ki})} \right\}. \quad (2.7)$$

The expansion point $\gamma_k^* = \arg \min_{\gamma} s_k(\gamma)$ is a solution to:

$$\mathbf{G}^{-1}\gamma - \mathbf{Z}'_k \mathbf{W}_k \Delta_k (\mathbf{y}_k - \boldsymbol{\mu}_k) = \mathbf{0}. \quad (2.8)$$

◆

Proof. Recall that $s_k(\gamma)$ was defined as:

$$s_k(\gamma) = \frac{1}{2} \gamma' \mathbf{G}^{-1} \gamma - \sum_{i=1}^{n_k} \ell_{ki}(\gamma).$$

Finding the gradient of $s_k(\gamma)$ with respect to γ :

$$\begin{aligned} \frac{\partial s_k}{\partial \gamma} &= \mathbf{G}^{-1} \gamma - \sum_{i=1}^{n_k} \frac{\partial \ell_{ki}}{\partial \mu_{ki}} \frac{\partial \mu_{ki}}{\partial \eta_{ki}} \frac{\partial \eta_{ki}}{\partial \gamma} \\ &= \mathbf{G}^{-1} \gamma - \sum_{i=1}^{n_k} \frac{y_{ki} - \mu_{ki}}{\phi\nu(\mu_{ki})} \frac{1}{\dot{g}(\mu_{ki})} \mathbf{z}_{ki}. \end{aligned}$$

■

Proposition 2.3. Under the canonical link, the Hessian of $s_k(\gamma)$ w.r.t. γ is:

$$\ddot{s}_k(\gamma) = \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'} = \mathbf{G}^{-1} + \mathbf{Z}'_k \mathbf{W}_k \mathbf{Z}_k.$$

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Proof. Since $\dot{g}(\mu_{ki})\nu(\mu_{ki}) = 1$ under the canonical link, the gradient of $s_k(\gamma)$ is:

$$\dot{s}_k(\gamma) = \mathbf{G}^{-1}\gamma - \sum_{i=1}^{n_k} \frac{y_{ki} - \mu_{ki}}{\phi} \mathbf{z}_{ki}$$

Taking the partial of $\dot{s}_k(\gamma)$ w.r.t. γ' :

$$\begin{aligned} \ddot{s}(\gamma) &= \frac{\partial}{\partial \gamma'} \left\{ \mathbf{G}^{-1}\gamma - \sum_{i=1}^{n_k} \frac{y_{ki} - \mu_{ki}}{\phi} \mathbf{z}_{ki} \right\} \\ &= \mathbf{G}^{-1} + \sum_{i=1}^{n_k} \frac{\mathbf{z}_{ki}}{\phi} \frac{\partial \mu_{ki}}{\partial \eta_{ki}} \frac{\partial \eta_{ki}}{\partial \gamma'} \\ &= \mathbf{G}^{-1} + \sum_{i=1}^{n_k} \frac{\mathbf{z}_{ki}}{\phi} \frac{1}{\dot{g}(\mu_{ki})} \mathbf{z}'_{ki} \\ &= \mathbf{G}^{-1} + \sum_{i=1}^{n_k} \mathbf{z}_{ki} \frac{1}{\phi \nu(\mu_{ki}) \dot{g}^2(\mu_{ki})} \mathbf{z}'_{ki}. \end{aligned}$$

■

Proposition 2.4. Under the canonical link, the Laplace likelihood is expressible as:

$$\ell_q(\beta, \alpha, \gamma^*) \propto \sum_{k=1}^K \sum_{i=1}^{n_k} \ell_{ki}(\gamma_k^*) - \frac{1}{2} \sum_{k=1}^K (\gamma_k^*)' \mathbf{G}^{-1} \gamma_k^* - \frac{1}{2} \sum_{k=1}^K \ln \det (\mathbf{I} + \mathbf{G} \mathbf{Z}'_k \mathbf{W}_k \mathbf{Z}_k). \quad (2.9)$$

◆

Proof. Recall that $s_k(\gamma)$ was defined as:

$$s_k(\gamma) = \frac{1}{2} \gamma' \mathbf{G}^{-1} \gamma - \sum_{i=1}^{n_k} \ell_{ki}(\gamma)$$

Writing out the approximate quasi likelihood:

$$\begin{aligned} \ell(\beta, \alpha) &\propto -\frac{K}{2} \ln \det(\mathbf{G}) - \sum_{k=1}^K s_k(\gamma_k^*) - \frac{1}{2} \sum_{k=1}^K \ln \det \left\{ \frac{\partial^2 s_k}{\partial \gamma \partial \gamma'}(\gamma_k^*) \right\} \\ &= -\frac{K}{2} \ln \det(\mathbf{G}) - \frac{1}{2} \sum_{k=1}^K (\gamma_k^*)' \mathbf{G}^{-1} \gamma_k^* + \sum_{k=1}^K \sum_{i=1}^{n_k} \ell_{ki}(\gamma_k^*) - \frac{1}{2} \sum_{k=1}^K \ln \det (\mathbf{G}^{-1} + \mathbf{Z}'_k \mathbf{W}_k \mathbf{Z}_k). \end{aligned}$$

Combining the log determinant terms gives the result. ■

2.4 Penalized Quasi Likelihood

Remark 2.4. In *penalized quasi likelihood* (PQL), the Laplace objective is simplified by supposing that the weight matrix $\mathbf{W}_k$ changes only slowly with respect to μ_{ki} . For fixed $\boldsymbol{\alpha}$, a simple iteratively reweighted least squares (IRLS) procedure is available for solving the PQL score equations for $\boldsymbol{\beta}$ and $\boldsymbol{\gamma}$. For additional details, see [2]. $\blacklozenge$

Definition 2.2. Suppose that the weight matrix in (2.7) changes slowly w.r.t. the conditional mean:

$$\frac{\partial \mathbf{W}_k}{\partial \mu_{ki}} \approx \mathbf{0}. \quad (\text{A1})$$

Under the PQL assumption (A1), the Laplace objective (2.9) is proportional to the **PQL objective** ℓ_{PQL} , which is *defined* as:

$$\ell_{\text{PQL}}(\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\gamma}) \propto \sum_{k=1}^K \sum_{i=1}^{n_k} \ell_{ki}(\boldsymbol{\beta}, \boldsymbol{\alpha}, \boldsymbol{\gamma}_k) - \frac{1}{2} \sum_{k=1}^K \boldsymbol{\gamma}_k' \mathbf{G}^{-1}(\boldsymbol{\alpha}) \boldsymbol{\gamma}_k. \quad (2.10)$$

In the PQL framework, $\boldsymbol{\gamma}_k$ is treated as a nuisance parameter that requires estimation. $\blacksquare$

Proposition 2.5 (PQL Score Equations). The PQL score equations are:

$$\begin{aligned} \mathcal{U}_{\boldsymbol{\beta}} &= \frac{\partial \ell_{\text{PQL}}}{\partial \boldsymbol{\beta}} = \sum_{k=1}^K \mathbf{X}_k' \mathbf{W}_k \boldsymbol{\Delta}_k (\mathbf{y}_k - \boldsymbol{\mu}_k), \\ \mathcal{U}_{\boldsymbol{\gamma}_k} &= \frac{\partial \ell_{\text{PQL}}}{\partial \boldsymbol{\gamma}_k} = -\mathbf{G}^{-1} \boldsymbol{\gamma}_k + \mathbf{Z}_k' \mathbf{W}_k \boldsymbol{\Delta}_k (\mathbf{y}_k - \boldsymbol{\mu}_k). \end{aligned}$$

$\blacklozenge$

Proof. The score for $\boldsymbol{\beta}$ is:

$$\mathcal{U}_{\boldsymbol{\beta}} = \frac{\partial \ell_{\text{PQL}}}{\partial \boldsymbol{\beta}} = \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{\partial \ell_{ki}}{\partial \mu_{ki}} \frac{\partial \mu_{ki}}{\partial \eta_{ki}} \frac{\partial \eta_{ki}}{\partial \boldsymbol{\beta}} = \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{y_{ki} - \mu_{ki}}{\phi \nu(\mu_{ki})} \frac{1}{\dot{g}(\mu_{ki})} \mathbf{x}_{ki}.$$

The score for $\boldsymbol{\gamma}_k$ was obtained in (2.8). $\blacksquare$

Proposition 2.6. Under the canonical link, the PQL information matrices are:

- For β :

$$\mathcal{J}_{\beta\beta'} = \sum_{k=1}^K \mathbf{X}_k' \mathbf{W}_k \mathbf{X}_k.$$

- For γ_k :

$$\mathcal{J}_{\gamma_k \gamma_k'} = \mathbf{G}^{-1} + \mathbf{Z}_k' \mathbf{W}_k \mathbf{Z}_k.$$

- Cross information:

$$\mathcal{J}_{\beta \gamma_k'} = \mathbf{X}_k' \mathbf{W}_k \mathbf{Z}_k.$$

◆

Proof. Recall that under the canonical link $\nu(\mu_{ki})\dot{g}(\mu_{ki}) = 1$. The PQL score for β simplifies to:

$$\mathcal{U}_\beta = \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{y_{ki} - \mu_{ki}}{\phi} \mathbf{x}_{ki}.$$

The Hessian w.r.t. β is:

$$\mathcal{H}_{\beta\beta'} = \frac{\partial \mathcal{U}_\beta}{\partial \beta'} = - \sum_{k=1}^K \sum_{i=1}^{n_k} \mathbf{x}_{ki} \frac{1}{\phi \dot{g}(\mu_{ki})} \mathbf{x}_{ki}' = - \sum_{k=1}^K \mathbf{X}_k' \mathbf{W}_k \mathbf{X}_k.$$

The PQL score equation for γ_k is:

$$\mathcal{U}_{\gamma_k} = -\mathbf{G}^{-1} \gamma_k + \sum_{i=1}^{n_k} \frac{y_{ki} - \mu_{ki}}{\phi} \mathbf{z}_{ki}$$

The Hessian w.r.t. γ_k is:

$$\mathcal{H}_{\gamma_k \gamma_k'} = \frac{\partial \mathcal{U}_{\gamma_k}}{\partial \gamma_k'} = -\mathbf{G}^{-1} - \sum_{i=1}^{n_k} \mathbf{z}_{ki} \frac{1}{\phi \dot{g}(\mu_{ki})} \mathbf{z}_{ki}' = -\mathbf{G}^{-1} - \mathbf{Z}_k' \mathbf{W}_k \mathbf{Z}_k.$$

Finally, the cross Hessian matrix is:

$$\mathcal{H}_{\beta \gamma_k'} = \frac{\partial \mathcal{U}_\beta}{\partial \gamma_k'} = - \sum_{i=1}^{n_k} \mathbf{x}_{ki} \frac{1}{\phi \dot{g}(\mu_{ki})} \mathbf{z}_{ki}' = -\mathbf{X}_k' \mathbf{W}_k \mathbf{Z}_k.$$

■

Estimation

3.1 Working Vector

Remark 3.1. Let $n = \sum_{k=1}^K n_k$. To simplify notation, define the following structures:

Structure	Dimension	Description
$\mathbf{y} = \text{vec}(\mathbf{y}_1, \dots, \mathbf{y}_K)$	$n \times 1$	Response vector.
$\mathbf{W} = \text{diag}(\mathbf{W}_1, \dots, \mathbf{W}_K)$	$n \times n$	Weight matrix.
$\mathbf{\Delta} = \text{diag}(\mathbf{\Delta}_1, \dots, \mathbf{\Delta}_K)$	$n \times n$	Delta matrix.
$\boldsymbol{\mu} = \text{vec}(\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_K)$	$n \times 1$	Mean vector.
$\mathbf{X} = \text{rbind}(\mathbf{X}_1, \dots, \mathbf{X}_K)$	$n \times p$	Fixed effect covariates.
$\boldsymbol{\gamma} = \text{vec}(\boldsymbol{\gamma}_1, \dots, \boldsymbol{\gamma}_K)$	$Kq \times 1$	Random effect.
$\mathbf{Z} = \text{diag}(\mathbf{Z}_1, \dots, \mathbf{Z}_K)$	$n \times Kq$	Random effect covariates.
$\mathcal{G}^{-1} = \mathbf{I}_{K \times K} \otimes \mathbf{G}^{-1}$	$Kq \times Kq$	Random effect covariance.

◆

Definition 3.1. Define the **working vector** $\tilde{\mathbf{y}}$ as:

$$\tilde{\mathbf{y}} = \boldsymbol{\eta} + \mathbf{\Delta}(\mathbf{y} - \boldsymbol{\mu}). \quad (3.1)$$

Here the linear predictor is $\boldsymbol{\eta} \equiv \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}$ and the mean vector is $\boldsymbol{\mu} = h(\boldsymbol{\eta})$. ■

3.2 Normal Equations

Proposition 3.1 (Fisher Scoring Algorithm). Fix the covariance parameters $\boldsymbol{\alpha}$. The solutions $\hat{\boldsymbol{\beta}}$ and $\hat{\boldsymbol{\gamma}}$ to the PQL score equations satisfy the GLMM **normal equations**:

$$\begin{pmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{Z} \\ \mathbf{Z}'\mathbf{W}\mathbf{X} & \mathbf{Z}'\mathbf{W}\mathbf{Z} + \mathcal{G}^{-1} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathbf{W}\tilde{\mathbf{y}} \\ \mathbf{Z}'\mathbf{W}\tilde{\mathbf{y}} \end{pmatrix}, \quad (3.2)$$

where $\tilde{\mathbf{y}}$ is the working vector (3.1). ◆

Proof. The PQL score equations are compactly expressible as:

$$\begin{aligned} \mathcal{U}_{\boldsymbol{\beta}} &= \mathbf{X}'\mathbf{W}\mathbf{\Delta}(\mathbf{y} - \boldsymbol{\mu}), \\ \mathcal{U}_{\boldsymbol{\gamma}} &= -\mathcal{G}^{-1}\boldsymbol{\gamma} + \mathbf{Z}'\mathbf{W}\mathbf{\Delta}(\mathbf{y} - \boldsymbol{\mu}). \end{aligned} \quad (3.3)$$

Let $\mathcal{U} = (\mathcal{U}_{\boldsymbol{\beta}}, \mathcal{U}_{\boldsymbol{\gamma}})'$ denote the stacked score equations, and let $\boldsymbol{\theta} = (\boldsymbol{\beta}, \boldsymbol{\gamma})$ group the parameters. Take the first order Taylor expansion of $\mathcal{U}$ around the PQL estimate $\hat{\boldsymbol{\theta}}$:

$$\mathbf{0} = \mathcal{U}(\hat{\boldsymbol{\theta}}) = \mathcal{U}(\boldsymbol{\theta}) + \mathcal{H}(\boldsymbol{\theta})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}) + \mathcal{O}_p(1).$$

Under the canonical link, the PQL Hessian is:

$$\mathcal{H}(\boldsymbol{\theta}) = \frac{\partial \mathcal{U}}{\partial \boldsymbol{\theta}'}(\boldsymbol{\theta}) = - \begin{pmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{Z} \\ \mathbf{Z}'\mathbf{W}\mathbf{X} & \mathbf{Z}'\mathbf{W}\mathbf{Z} + \mathcal{G}^{-1} \end{pmatrix}.$$

From the first order Taylor expansion:

$$\begin{aligned} \mathcal{U}(\boldsymbol{\theta}) &\approx -\mathcal{H}(\boldsymbol{\theta})(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}), \\ \begin{pmatrix} \mathbf{X}'\mathbf{W}\Delta(\mathbf{y} - \boldsymbol{\mu}) \\ \mathbf{Z}'\mathbf{W}\Delta(\mathbf{y} - \boldsymbol{\mu}) - \mathcal{G}^{-1}\boldsymbol{\gamma} \end{pmatrix} &\approx \begin{pmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{Z} \\ \mathbf{Z}'\mathbf{W}\mathbf{X} & \mathbf{Z}'\mathbf{W}\mathbf{Z} + \mathcal{G}^{-1} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \\ \hat{\boldsymbol{\gamma}} - \boldsymbol{\gamma} \end{pmatrix} \end{aligned}$$

Rearranging gives the PQL *normal equations*:

$$\begin{pmatrix} \mathbf{X}'\mathbf{W}\mathbf{X} & \mathbf{X}'\mathbf{W}\mathbf{Z} \\ \mathbf{Z}'\mathbf{W}\mathbf{X} & \mathbf{Z}'\mathbf{W}\mathbf{Z} + \mathcal{G}^{-1} \end{pmatrix} \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\boldsymbol{\gamma}} \end{pmatrix} = \begin{pmatrix} \mathbf{X}'\mathbf{W}\tilde{\mathbf{y}} \\ \mathbf{Z}'\mathbf{W}\tilde{\mathbf{y}} \end{pmatrix},$$

where $\tilde{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \Delta(\mathbf{y} - \boldsymbol{\mu}) = \boldsymbol{\eta} + \Delta(\mathbf{y} - \boldsymbol{\mu})$ is the working vector. ■

Remark 3.2. Comparing the PQL normal equations (3.2) with *mixed model equations* (see notes on LMMs) demonstrates that the working-vector $\tilde{\mathbf{y}}$ follows a LMM:

$$\begin{aligned} \tilde{\mathbf{y}} &= \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\epsilon} \\ \boldsymbol{\gamma} &\sim N(\mathbf{0}, \mathcal{G}) \perp \boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{W}^{-1}) \end{aligned} \tag{3.4}$$

The induced marginal model is:

$$\tilde{\mathbf{y}} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

where $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \boldsymbol{\Sigma})$ with:

$$\boldsymbol{\Sigma} = \mathbf{W}^{-1} + \mathbf{Z}\mathcal{G}\mathbf{Z}'. \tag{3.5}$$

From the results for LMMs, the best linear unbiased estimator of $\boldsymbol{\beta}$ is:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1}\tilde{\mathbf{y}}. \tag{3.6}$$

Define the error projection:

$$\mathbf{Q} = \boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1}.$$

The difference between the working response $\tilde{\mathbf{y}}$ and $\mathbf{X}\hat{\boldsymbol{\beta}}$ is expressible as:

$$\tilde{\mathbf{y}} - \mathbf{X}\hat{\boldsymbol{\beta}} = \boldsymbol{\Sigma}\{\boldsymbol{\Sigma}^{-1} - \boldsymbol{\Sigma}^{-1}\mathbf{X}(\mathbf{X}'\boldsymbol{\Sigma}^{-1}\mathbf{X})^{-1}\mathbf{X}'\boldsymbol{\Sigma}^{-1}\}\tilde{\mathbf{y}} = \boldsymbol{\Sigma}\mathbf{Q}\tilde{\mathbf{y}}.$$

The best linear predictor of $\boldsymbol{\gamma}$ is:

$$\hat{\boldsymbol{\gamma}} = \hat{\mathbb{E}}(\boldsymbol{\gamma}|\tilde{\mathbf{y}}) = \mathcal{G}\mathbf{Z}'\boldsymbol{\Sigma}^{-1}(\tilde{\mathbf{y}} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathcal{G}\mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}}. \tag{3.7}$$

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3.3 Variance Component Estimation

Remark 3.3. For fixed α , the PQL normal equations (3.2) suggest the following procedure for estimating β and γ .

Algorithm 1 PQL Estimation

Require: Covariance parameters α ; observed data $\mathcal{D} = \{\mathbf{y}, \mathbf{X}, \mathbf{Z}\}$.

- 1: Initialize $\beta^{(0)}$ and $\gamma^{(0)}$.
 - 2: **repeat**
 - 3: Using the current $(\beta^{(r)}, \gamma^{(r)})$, construct:
 - 4: **substeps**
 - 5: $\Delta^{(r)} = \text{diag}\{\dot{g}(\mu_{ki}^{(r)})\}$.
 - 6: $\mathbf{W}^{(r)} = \text{diag}\{\phi\nu(\mu_{ki}^{(r)})\dot{g}^2(\mu_{ki}^{(r)})\}^{-1}$.
 - 7: $\tilde{\mathbf{y}}^{(r)} = \boldsymbol{\eta}^{(r)} + \Delta^{(r)}(\mathbf{y} - \boldsymbol{\mu}^{(r)})$.
 - 8: $\Sigma^{(r)} = \mathbf{W}^{(r),-1} + \mathbf{Z}\mathcal{G}\mathbf{Z}'$.
 - 9: **end substeps**
 - 10: Update: $\beta^{(r+1)} \leftarrow (\mathbf{X}'\Sigma^{(r),-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{(r),-1}\tilde{\mathbf{y}}^{(r)}$.
 - 11: Update: $\gamma^{(r+1)} \leftarrow \mathcal{G}\mathbf{Z}'\Sigma^{(r),-1}(\tilde{\mathbf{y}} - \mathbf{X}\beta^{(r+1)})$.
 - 12: **until** $(\beta^{(r+1)}, \gamma^{(r+1)})$ stabilize.
-

Towards obtaining an estimation procedure for α , the PQL estimates of β and γ are substituted into the PQL objective (2.10) to obtain a *profile objective* for the variance components. ◆

3.3.1 Total Quasi Likelihood

Proposition 3.2. The difference between the working vector $\tilde{\mathbf{y}}$ and the fitted linear predictor $\hat{\boldsymbol{\eta}}$ is expressible as:

$$\tilde{\mathbf{y}} - \hat{\boldsymbol{\eta}} = \mathbf{W}^{-1}\mathbf{Q}\tilde{\mathbf{y}}. \quad (3.8)$$
◆

Proof. The fitted linear predictor is:

$$\hat{\boldsymbol{\eta}} = \mathbf{X}\hat{\boldsymbol{\beta}} + \mathbf{Z}\hat{\boldsymbol{\gamma}}.$$

Recall that $\tilde{\mathbf{y}} - \mathbf{X}\hat{\boldsymbol{\beta}} = \Sigma\mathbf{Q}\tilde{\mathbf{y}}$ and that $\mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{Z}\mathcal{G}\mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}}$, thus:

$$\tilde{\mathbf{y}} - \hat{\boldsymbol{\eta}} = (\tilde{\mathbf{y}} - \mathbf{X}\hat{\boldsymbol{\beta}}) - \mathbf{Z}\hat{\boldsymbol{\gamma}} = \Sigma\mathbf{Q}\tilde{\mathbf{y}} - \mathbf{Z}\mathcal{G}\mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}} = (\Sigma - \mathbf{Z}\mathcal{G}\mathbf{Z}')\mathbf{Q}\tilde{\mathbf{y}}.$$

The conclusion follows since $\Sigma = \mathbf{W}^{-1} + \mathbf{Z}\mathcal{G}\mathbf{Z}'$, or $\Sigma - \mathbf{Z}\mathcal{G}\mathbf{Z}' = \mathbf{W}^{-1}$. ■

Definition 3.2. Let T denote the *total quasi (log) likelihood*:

$$T(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}_k) \equiv \sum_{k=1}^K \sum_{i=1}^{n_k} \ell_{ki} = \sum_{k=1}^K \sum_{i=1}^{n_k} \int_{y_{ki}}^{\mu_{ki}} \frac{(y_{ki} - u)}{\phi \nu(u)} du.$$

The **quasi deviance** is -2ϕ times the total quasi likelihood:

$$D(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}_k) \equiv -2\phi \cdot T = -2 \sum_{k=1}^K \sum_{i=1}^{n_k} \int_{y_{ki}}^{\mu_{ki}} \frac{(y_{ki} - u)}{\nu(u)} du.$$

The Pearson χ^2 approximation to the model deviance is:

$$D \approx \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{(y_{ki} - \hat{\mu}_{ki})^2}{\nu(\hat{\mu}_{ki})}.$$

■

Proposition 3.3. The Pearson approximation to the total quasi likelihood is:

$$\tilde{T} = -\frac{1}{2} \tilde{\mathbf{y}}' \mathbf{Q} \mathbf{W}^{-1} \mathbf{Q} \tilde{\mathbf{y}}. \quad (3.9)$$

◆

Proof. Using the Pearson χ^2 approximation to the deviance :

$$D = -2\phi \cdot T \approx \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{(y_{ki} - \hat{\mu}_{ki})^2}{\nu(\hat{\mu}_{ki})}.$$

Define the Pearson approximation to the total quasi likelihood as:

$$\tilde{T} = -\frac{1}{2} \sum_{k=1}^K \sum_{i=1}^{n_k} \frac{(y_{ki} - \hat{\mu}_{ki})^2}{\phi \nu(\hat{\mu}_{ki})}.$$

Expressed in matrix notation:

$$\tilde{T} = -\frac{1}{2} \sum_{k=1}^K (\mathbf{y}_k - \hat{\boldsymbol{\mu}}_k)' \boldsymbol{\Delta}_k \mathbf{W}_k \boldsymbol{\Delta}_k (\mathbf{y}_k - \hat{\boldsymbol{\mu}}_k) = -\frac{1}{2} (\mathbf{y} - \hat{\boldsymbol{\mu}})' \boldsymbol{\Delta} \mathbf{W} \boldsymbol{\Delta} (\mathbf{y} - \hat{\boldsymbol{\mu}}).$$

Recall that the working vector was defined as $\tilde{\mathbf{y}} = \hat{\boldsymbol{\eta}} + \boldsymbol{\Delta}(\mathbf{y} - \hat{\boldsymbol{\mu}})$. Replacing $\boldsymbol{\Delta}(\mathbf{y} - \hat{\boldsymbol{\mu}})$ by $(\tilde{\mathbf{y}} - \hat{\boldsymbol{\eta}})$ gives $\tilde{T} = -\frac{1}{2}(\tilde{\mathbf{y}} - \hat{\boldsymbol{\eta}})' \mathbf{W}(\tilde{\mathbf{y}} - \hat{\boldsymbol{\eta}})$. Finally, applying (3.8):

$$\tilde{T} = -\frac{1}{2} \tilde{\mathbf{y}}' \mathbf{Q} \mathbf{W}^{-1} \mathbf{Q} \tilde{\mathbf{y}}.$$

■

3.3.2 Sylvester's Identity

Proposition 3.4. Consider the matrices $\mathbf{A}_{n \times q}$ and $\mathbf{B}_{q \times n}$ with $q < n$. The matrices $(\mathbf{AB})_{n \times n}$ and $(\mathbf{BA})_{q \times q}$ have the same non-zero eigenvalues. $\blacklozenge$

Proof. Suppose $\lambda \neq 0$ is an eigenvalue of $\mathbf{BA}$. There exists an eigenvector $\mathbf{u}_\lambda$, distinct from the null vector $\mathbf{u}_\lambda \neq \mathbf{0}$, s.t. $\mathbf{BAu}_\lambda = \lambda \mathbf{u}_\lambda$. Left multiply by $\mathbf{A}$ to obtain:

$$\mathbf{ABAu}_\lambda = \lambda \mathbf{Au}_\lambda \quad \mathbf{AB}(\mathbf{Au}_\lambda) = \lambda(\mathbf{Au}_\lambda).$$

Thus, $\lambda \neq 0$ is an eigenvalue of $\mathbf{BA}$. Analogous reasoning shows that if $\lambda \neq 0$ is an eigenvalue of $\mathbf{AB}$, then $\lambda \neq 0$ is an eigenvalue of $\mathbf{BA}$. $\blacksquare$

Proposition 3.5 (Sylvester). Consider the $\mathbf{A}_{n \times q}$ and $\mathbf{B}_{q \times n}$ with $q < n$, then:

$$\det(\mathbf{I}_{n \times n} + \mathbf{AB}) = \det(\mathbf{I}_{q \times q} + \mathbf{BA}). \quad (3.10)$$

Proof. Let $\mathbf{BA} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$ denote the eigenvalue decomposition of the $q \times q$ matrix $\mathbf{BA}$. Expressing the determinant of $\mathbf{I} + \mathbf{BA}$ as a product of eigenvalues:

$$\det(\mathbf{I} + \mathbf{BA}) = \det(\mathbf{I} + \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}) = \det\{\mathbf{U}(\mathbf{I} + \mathbf{\Lambda})\mathbf{U}^{-1}\} = \det(\mathbf{I} + \mathbf{\Lambda}) = \prod_{j=1}^q (1 + \lambda_j).$$

Observe that the set of null eigenvalues $\{j : \lambda_j = 0\}$ contribute a one to the product, and hence leave it unchanged. Thus:

$$\det(\mathbf{I} + \mathbf{BA}) = \prod_{\{j: \lambda_j \neq 0\}} (1 + \lambda_j).$$

Now let $\mathbf{AB} = \mathbf{V}\mathbf{T}\mathbf{V}^{-1}$ denote the eigenvalue decomposition of the $n \times n$ matrix $\mathbf{AB}$. $\det(\mathbf{I} + \mathbf{AB})$ is likewise expressible as a product over the non-zero eigenvalues of $\mathbf{AB}$:

$$\det(\mathbf{I} + \mathbf{AB}) = \prod_{\{j: \tau_j \neq 0\}} (1 + \tau_j)$$

Since $\mathbf{AB}$ and $\mathbf{BA}$ have the same non-zero eigenvalues, conclude that:

$$\det(\mathbf{I} + \mathbf{AB}) = \prod_{\{j: \tau_j \neq 0\}} (1 + \tau_j) = \prod_{\{j: \lambda_j \neq 0\}} (1 + \lambda_j) = \det(\mathbf{I} + \mathbf{BA}).$$

$\blacksquare$

Proposition 3.6. Under the PQL assumption (A1):

$$\ln \det(\mathbf{I} + \mathbf{GZ}'_k \mathbf{W}_k \mathbf{Z}_k) \propto \ln \det(\mathbf{\Sigma}_k). \quad (3.11)$$

$\blacklozenge$

Proof. Let $\mathbf{B} = \mathbf{G}\mathbf{Z}'_k$ and $\mathbf{A} = \mathbf{W}_k\mathbf{Z}_k$. Using Sylvester's identity (3.10):

$$\ln \det(\mathbf{I} + \mathbf{G}\mathbf{Z}'_k\mathbf{W}_k\mathbf{Z}_k) = \ln \det(\mathbf{I} + \mathbf{W}_k\mathbf{Z}_k\mathbf{G}\mathbf{Z}'_k).$$

Recall from the working vector LMM (3.4) that:

$$\mathbb{V}(\tilde{\mathbf{y}}_k | \mathbf{X}_k, \mathbf{Z}_k) = \mathbf{W}_k^{-1} + \mathbf{Z}_k\mathbf{G}\mathbf{Z}'_k.$$

Expressing the log determinant in terms of Σ_k :

$$\ln \det(\mathbf{I} + \mathbf{W}_k\mathbf{Z}_k\mathbf{G}\mathbf{Z}'_k) = \ln \det(\mathbf{W}_k\Sigma_k) = \ln \det(\mathbf{W}_k) + \ln \det(\Sigma_k).$$

Now, assuming $\mathbf{W}_k$ changes only slowly w.r.t. μ_{ki} , such that $\partial_\alpha \ln \det(\mathbf{W}_k) \approx 0$:

$$\ln \det(\mathbf{I} + \mathbf{G}\mathbf{Z}'_k\mathbf{W}_k\mathbf{Z}_k) \propto \ln \det(\Sigma_k).$$

■

3.3.3 Profile Quasi Likelihood

Result 3.1. Under the PQL assumption (A1), a profile quasi likelihood for the variance components α is:

$$\ell_{pq}(\alpha) \propto -\frac{1}{2} \ln \det(\Sigma) - \frac{1}{2} \tilde{\mathbf{y}}' \mathbf{Q} \tilde{\mathbf{y}}, \quad (3.12)$$

where $\Sigma = \mathbf{W}^{-1} + \mathbf{Z}\mathbf{G}\mathbf{Z}'$ and $\mathbf{Q} = \Sigma^{-1} - \Sigma^{-1}\mathbf{X}(\mathbf{X}'\Sigma^{-1}\mathbf{X})^{-1}\mathbf{X}'\Sigma^{-1}$ are defined as for the working vector LMM (3.4). ♣

Proof. Recall that the Laplace objective (2.9) took the form:

$$\ell_q(\beta, \alpha, \gamma) \propto \sum_{k=1}^K \sum_{i=1}^{n_k} \ell_{ki} - \frac{1}{2} \sum_{k=1}^K (\gamma_k)' \mathbf{G}^{-1} \gamma_k - \frac{1}{2} \sum_{k=1}^K \ln \det(\mathbf{I} + \mathbf{G}\mathbf{Z}'_k\mathbf{W}_k\mathbf{Z}_k).$$

Proceed by substituting the PQL estimates of β (3.6) and of γ (3.7) into the Laplace objective to form a profile objective. The leading term is the total quasi likelihood T , which may be approximated as (3.9) using Pearson's statistic:

$$\sum_{k=1}^K \sum_{i=1}^{n_k} \ell_{ki}(\hat{\gamma}_k) \approx -\frac{1}{2} \tilde{\mathbf{y}}' \mathbf{Q} \mathbf{W}^{-1} \mathbf{Q} \tilde{\mathbf{y}}.$$

The second term, evaluated at the BLUP $\hat{\gamma} = \mathbf{G}\mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}}$, is expressible as:

$$-\frac{1}{2} \sum_{k=1}^K \hat{\gamma}'_k \mathbf{G}^{-1} \hat{\gamma}_k = -\frac{1}{2} \hat{\gamma}' \mathbf{G}^{-1} \hat{\gamma} = -\frac{1}{2} \tilde{\mathbf{y}}' \mathbf{Q} \mathbf{Z} \mathbf{G} \mathbf{G}^{-1} \mathbf{G} \mathbf{Z}' \mathbf{Q} \tilde{\mathbf{y}}.$$

Combining the first and second terms gives:

$$-\frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\mathbf{W}^{-1}\mathbf{Q}\tilde{\mathbf{y}} - \frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\mathbf{Z}\mathbf{G}\mathbf{Z}'\mathbf{Q}\tilde{\mathbf{y}} = -\frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}(\mathbf{W}^{-1} + \mathbf{Z}\mathbf{G}\mathbf{Z}')\mathbf{Q}\tilde{\mathbf{y}} = -\frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\mathbf{\Sigma}\mathbf{Q}\tilde{\mathbf{y}},$$

where $\mathbf{\Sigma}$ is the working vector covariance (3.5). Direct calculation demonstrates that $\mathbf{Q}\mathbf{\Sigma}\mathbf{Q} = \mathbf{Q}$ (see notes on ReML). Therefore:

$$\ell_q(\hat{\boldsymbol{\beta}}, \boldsymbol{\alpha}, \hat{\boldsymbol{\gamma}}) = -\frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\tilde{\mathbf{y}} - \frac{1}{2}\sum_{k=1}^K \ln \det(\mathbf{I} + \mathbf{G}\mathbf{Z}'_k\mathbf{W}_k\mathbf{Z}_k).$$

By proportion (3.11), the third term is proportionate to:

$$-\frac{1}{2}\sum_{k=1}^K \ln \det(\mathbf{I} + \mathbf{G}\mathbf{Z}'_k\mathbf{W}_k\mathbf{Z}_k) \propto -\frac{1}{2}\sum_{k=1}^K \ln \det(\mathbf{\Sigma}_k) = -\frac{1}{2}\ln \det(\mathbf{\Sigma}).$$

Overall, the profile log likelihood for $\boldsymbol{\alpha}$ is:

$$\begin{aligned} \ell_{pq}(\boldsymbol{\alpha}) &\equiv \ell_q(\hat{\boldsymbol{\beta}}, \boldsymbol{\alpha}, \hat{\boldsymbol{\gamma}}) = -\frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\tilde{\mathbf{y}} - \frac{1}{2}\ln \det(\mathbf{\Sigma}) \\ &= -\frac{1}{2}\sum_{k=1}^K \{\tilde{\mathbf{y}}'_k\mathbf{Q}_k\tilde{\mathbf{y}}_k + \ln \det(\mathbf{\Sigma}_k)\}. \end{aligned}$$

■

Remark 3.4. A restricted profile quasi likelihood is formed by applying a Jeffrey's prior $\pi(\boldsymbol{\beta}) \propto \det(\mathcal{I}_{\boldsymbol{\beta}\boldsymbol{\beta}'})^{-1/2} = \det(\mathbf{X}'\mathbf{\Sigma}^{-1}\mathbf{X})^{-1/2}$ to the fixed effects to obtain:

$$\ell_{rpq}(\boldsymbol{\alpha}) \propto \ell_{pq}(\boldsymbol{\alpha}) + \ln \pi(\boldsymbol{\beta}) = -\frac{1}{2}\ln \det(\mathbf{\Sigma}) - \frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\tilde{\mathbf{y}} - \frac{1}{2}\ln \det(\mathbf{X}'\mathbf{\Sigma}^{-1}\mathbf{X}). \quad (3.13)$$

The ReML score equation for a variance component α_p is:

$$\frac{\partial \ell_{rpq}}{\partial \alpha_p} = -\frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial \mathbf{\Sigma}}{\partial \alpha_p}\right) + \frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\frac{\partial \mathbf{\Sigma}}{\partial \alpha_p}\mathbf{Q}\tilde{\mathbf{y}}.$$

The ReML cross information between α_p and α_q is:

$$\mathcal{I}_{\alpha_p\alpha_q} = \frac{1}{2}\text{tr}\left(\mathbf{Q}\frac{\partial \mathbf{\Sigma}}{\partial \alpha_p}\mathbf{Q}\frac{\partial \mathbf{\Sigma}}{\partial \alpha_q}\right).$$

Typically, the ReML score equations have no closed form and must be solved numerically. When this is so, a computationally efficient approximation to the ReML information matrix is the **average information matrix**:

$$\mathcal{A}_{\alpha_p\alpha_q} = \frac{1}{2}\tilde{\mathbf{y}}'\mathbf{Q}\frac{\partial \mathbf{\Sigma}}{\partial \alpha_p}\mathbf{Q}\frac{\partial \mathbf{\Sigma}}{\partial \alpha_q}\mathbf{Q}\tilde{\mathbf{y}}.$$

◆

Inference

4.1 Fixed Effects

Proposition 4.1. Consider the GLMM:

$$g(\mu_{ki}) = \mathbf{x}'_{A,ki} \boldsymbol{\beta}_A + \mathbf{x}'_{B,ki} \boldsymbol{\beta}_B + \mathbf{z}'_{ki} \boldsymbol{\gamma}_k, \quad (4.1)$$

where the fixed effect $\boldsymbol{\beta}$ has been partitioned into those parameters $\boldsymbol{\beta}_A$ that are restricted under the null, and those parameters $\boldsymbol{\beta}_B$ that are not. A score test of $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_{A0}$ is:

$$T_S = \tilde{\mathbf{y}}'_0 \tilde{\mathbf{Q}}_0 \mathbf{X}_A (\mathbf{X}'_A \tilde{\mathbf{Q}}_0 \mathbf{X}_A)^{-1} \mathbf{X}'_A \tilde{\mathbf{Q}}_0 \tilde{\mathbf{y}} \sim \chi^2_{\dim(\boldsymbol{\beta}_A)}.$$

The components of this statistic are:

- i. Null model working response:

$$\tilde{\mathbf{y}}_0 = \tilde{\boldsymbol{\eta}}_0 + \tilde{\boldsymbol{\Delta}}(\mathbf{y} - \tilde{\boldsymbol{\mu}}_0),$$

where $\tilde{\boldsymbol{\eta}}_0 = \mathbf{X}_A \boldsymbol{\beta}_{A0} + \mathbf{X}_B \tilde{\boldsymbol{\beta}}_B + \mathbf{Z} \tilde{\boldsymbol{\gamma}}$, $\tilde{\boldsymbol{\mu}}_0 = h(\tilde{\boldsymbol{\eta}}_0)$, and $(\tilde{\boldsymbol{\beta}}_B, \tilde{\boldsymbol{\gamma}})$ are estimates obtained by fitting (4.1) under $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_{A0}$.

- ii. Null model error projection:

$$\tilde{\mathbf{Q}}_0 = \tilde{\boldsymbol{\Sigma}}_0^{-1} - \tilde{\boldsymbol{\Sigma}}_0^{-1} \mathbf{X}_B (\mathbf{X}'_B \tilde{\boldsymbol{\Sigma}}_0^{-1} \mathbf{X}_B)^{-1} \mathbf{X}'_B \tilde{\boldsymbol{\Sigma}}_0^{-1},$$

where $\tilde{\boldsymbol{\Sigma}}_0 = \tilde{\mathbf{W}}_0^{-1} + \mathbf{Z} \tilde{\mathbf{G}} \mathbf{Z}'$ is the working vector covariance under $H_0 : \boldsymbol{\beta}_A = \boldsymbol{\beta}_{A0}$.

◆

Proof. From (3.3), the PQL score vector for $\boldsymbol{\beta}_A$ is:

$$\mathcal{U}_{\boldsymbol{\beta}_A} = \mathbf{X}'_A \mathbf{W} \boldsymbol{\Delta}(\mathbf{y} - \boldsymbol{\mu}_0) = \mathbf{X}'_A \mathbf{Q}_0 \tilde{\mathbf{y}}_0.$$

By direct calculation, the variance of the PQL score is:

$$\mathbb{V}(\mathcal{U}_{\boldsymbol{\beta}_A} | \mathbf{X}, \mathbf{Z}) = \mathbf{X}'_A \mathbf{Q}_0 \boldsymbol{\Sigma}_0 \mathbf{Q}_0 \mathbf{X}_A = \mathbf{X}'_A \mathbf{Q}_0 \mathbf{X}_A.$$

■

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